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Spring Core and MVC
Duration: 40 Hours

TOC version: 1.0

Day No	Session No	Learning Time (in hours)	Module Name	Topic	Sub-Topics	Remarks
Day 01	Session 01	0.5	Spring- Core and JDBC	Project Briefing	**Quick Overview -** **Agenda & Objective**: Apply Spring Core concepts in a real-world scenario. **Activities:** a. Present a business scenario (e.g., a simple real-time problem like e-comm app) b. Break down requirements and design the application architecture c. Walkthrough of design and application architecture d. Discussion on Technical Pre-requisites	Faculty to give an overview and explain the important aspects of these topics from implementation perspective through hands-on
		0.5		Project Driven: Spring Core	**Quick Overview -** Introduction to Spring Core and Setting Up a Spring Project	
		3			**Project Task :** Creation of Standalone App (with implicit concept explanation) **Activities :** a. Create a new Maven Project b. Include and Manage Spring Core Dependencies c. Illustrate Spring Bean Configuration in a JAVA Configuration file d. Implement the approach for creating a Spring Stand Alone Application using Annotation Based Spring IoC container(AnnotationConfigApplicationContext) e. Implement the Basic Stereotype Annotations (@Component, @Service, @Repository, @ComponentScan etc.,) f. Implement Dependency Injection & Scopes	
	Session 02	0.34		Project Driven: Spring - JDBC & Advantages	**Quick Overview -** Introduction to Spring JDBC, JDBC vs Spring JDBC	
		2			**Project Task :** Leveraging Spring JDBC and Integration **Objective:** Enhance the application with database functionalities leveraging Spring JDBC. **Activities:** a. Extend the application to include database interactions b. Implement features (eg., user registration, product listing)	
		1.66			**Project Task :** Database Interaction Extending the Application **Activities:** a. Creating bean for JdbcTemplate b. Injecting Datasource into JDBCTemplate Bean c. Running DDL : Statements using using the JdbCTemplate methods d. Updating DB: Leveraging the methods to Update the Database - Insert, Update, Delete e. Fetch: Executing Queries for Fetching data from DB (d and e : CRUD Operations)	
Day 02	Session 01	1	Spring MVC	Project Briefing	**Quick Overview:** **Agenda & Objective**: Setting the context of Leveraging Spring Framework Web Module for developing MVC-based Web Applications **Project Overview & Activities:** a. Present Business scenario - Building Web App (e.g., a simple real-time problem) b. Functional Requirements break down c. Discussion on Technical Pre-requisites d. Walkthrough of MVC Architecture and Design	Faculty to give an overview and explain the important aspects of these topics from implementation perspective through hands-on
		0.5		Spring MVC Architecture	**Quick Overview:** On Spring Web Module and how MVC Pattern is leveraged with the architecture and the famework components involved	
		1			**Project Task**: Architecture Discussion **Activities:** a. Detailed discussion on Spring MVC Architecture b. Understanding the Spring MVC Framework Components - DispatcherServlet, Controllers, Models, c. View Resolvers, XML with Bean Definitions d. Creating a flow diagram to illustrate request control and response generation e. Design and identify the files needed for respective Modules	
		0.5		Spring MVC Project Set Up	**Quick Overview:** How to set up Spring Project, Configuration of Dependencies and Directory structure resembling real time scenarios, Role of DispatcherServlet	
		1			**Project Task**: Setting up the Spring MVC project **Activities:** a. Create a new Maven project b. Configure dependencies for Spring Web c. Structure the project d. Configure DispatcherServlet	
	Session 02	0.5		Spring MVC Sample Flow	**Quick Overview:** Setting up the basic flow of a Spring MVC project involving the different technical components	
		0.5			**Project Task**: Executing a sample Spring MVC flow with Controller and JSP **Activities:** a. Setting up Spring MVC configurations b. Importance of WebApplicationContext c. Configuring View Resolver d. Creating a basic controller and JSP for display e. Running and testing the application	
		0.5		RequestMapping	**Quick Overview:** RequestMapping, Http Methods, Http Responses	
		1			**Project Task:** Ways to Handle Requests **Activities:** a. Implementing different URL patterns - Class Level, Method level, Both b. Mapping HTTP methods with RequestMapping c. Handling ambiguity and conflicts in RequestMapping s. Best practices when using RequestMapping	
		0.5			**Quick Overview:** Controller and Its importance, Annotations used for Controller creation, The different ways how a Controller receives the data from the request, Also the Spring Web Module provided Framework Components	

		1		Controller	**Project Task**: Building robust controllers **Activities:** a. Creating basic sample controllers and request handler methods b. Extracting Data from the request - @RequestParam, @PathVariable, @ModelAttribute c. Leveraging Spring components Model and ModelAndView for data exchange between views and controllers d. As a follow up, trainees to create all the needed Controllers for the Project	
Day 03	Session 01	0.5		Exception Handling	**Quick Overview:** Introduction to Spring Web Exception Handling and the Annotations involved	Faculty to give an overview and explain the important aspects of these topics from implementation perspective through hands-on
		1.5			**Project Task**: Exception Handling **Activities:** a. Identify the Technical and Business Exceptions needed to be incorporated b. Implement logic for - Handling exceptions with @ExceptionHandler and @ControllerAdvice c. Custom exception handling (optional)	
		0.5		Spring Forms	**Quick Overview:** Features of the Spring Forms and how they can be leveraged and come in handy for Data validation and Error Display	
		1.5			**Project Task:** Implementation of Spring Forms **Activities:** a. Develop a sample form using Spring taglib for one of the user registration form b. Simplified data binding using @ModelAttribute c. Form submission handling with @PostMapping d. Using Spring's Form Tag Library	
	Session 02	0.5		Data Validation and Binding	**Quick Overview:** Importance of Data Validation, Configuring the needed dependencies and leveraging the annotations for the Data validations	
		1.5			**Project Task**: Identify and implement the Business Validations specified **Activities:** a. Implementing form validation using Binding Result and b. Server-side validation using annotations like @Valid, @NotNull, @Size c. Handling validation errors and displaying appropriate error messages d. Validating nested objects	
		0.5		Spring JDBC Integration	**Quick Overview:** Integrating the Spring JDBC for demoing the persistence of the Data in the Database	
		1.5			**Project Task:** Integration of Spring JDBC with the created Spring MVC Web Application, which provides a storage component as well to store and retrieve data from the database	
Day 04	Session 01	4	Assignment	Assignment + Track-End Quiz Assessment		
	Session 02	4	Assignment	Assignment + Track-End Quiz Assessment		
Day 05	Session 01	4	Assignment	Assignment + Track-End Quiz Assessment		
	Session 02	4	Assignment	Assignment + Track-End Quiz Assessment		